Evaluating Factors that Influence Standardized Test Performance for Middle Grade Students in Virginia Over Ten Years

Jennifer Reif
George Mason University
AIT 580-011: Analytics: Big Data to Information
October 2025

Abstract

Since the introduction of the No Child Left Behind Act of 2001 there has been an increased emphasis on standardized testing as a measure of school performance. Virginia assesses its middle grade (Grades 6 – 8) students annually with Mathematics and English:Reading exams. In addition, 8th graders are tested in Science, English:Writing, and History and Social Science. This study analyzed test scores from 2015 through 2024 and evaluated the impacts of different funding sources, budget priorities, and the impact of the COVID-19 pandemic on student test performance. The pandemic caused significant declines in test performance, where most school districts saw their post-pandemic exam pass rates fall by an average of 20 points when testing resumed in 2021. In subsequent years, many districts have reported improving scores, while still lagging the pre-pandemic averages. This study found slight but statistically significant correlations between the source of school funding (local, state, or federal money). It did not find a significant relationship between budget priorities and student test performance.

Introduction

Virginia is a diverse state with over 130 school districts. Districts vary in geographical size from the City of Falls Church outside Washington, D.C. which covers approximately 2 square miles to Pittsylvania County that covers nearly 1,000 square miles in southwestern Virginia. Student populations also vary dramatically in different regions of the state. In 2024, Highland County had a total of 196 students across all grades from Kindergarten through Grade 12. By comparison, Fairfax County was the most populous school district with 176,402 students in the same year [1]. The number of middle schools in each district mirrors the differences in student population numbers. In 2024, 47 Virginia school districts had one school for covering the middle grades. By comparison, Fairfax County educated middle grade students in 187 different schools.

In Virginia, the Virginia Department of Education established the Standards of Learning (SOL) to outline the minimum expectations for student learning and achievement in English, Mathematics, Science, and History and Social Studies. The SOL tests administered in the spring of each year measure students' success in meeting those goals [2]. These tests are also used to determine the accreditation status of schools and meet state and federal reporting requirements [3]. Student testing performance is scored on a scale of 0 – 600. Scores of 0 – 399 are assigned a "FAIL" performance level while scores that are 400 – 499 are assigned a "Pass" or "Pass/Proficient" performance level. Scores above 500 are assigned to the "Pass/Advanced" performance level [4].

Since the introduction of the *No Child Left Behind Act of 2001*, there has been an increased emphasis on standardized testing in educational environments. The *American Recovery and Reinvestment Act of 2009* included additional investments in education and more reliance on standardized tests. Under *Every Student Succeeds Act* (ESSA Federal Legislation) of 2015, school districts are required to test students' performance in reading and mathematics annually in Grades 3 – 8. This increased focus on standardized tests has caused them to become a benchmark of education quality in specific schools and school districts [5].

While there are many complex factors that contribute to student success, school funding is quantifiable and lends itself to analysis. School funding in Virginia is governed by a complex system of policies and regulations and the local, state, and federal levels. It is important to identify how funding from different sources and spent on competing budget priorities correlate with student success in order to be good stewards of the limited public funding available.

In early 2020 the COVID-19 pandemic caused unprecedented disruptions to schools across the state. In response to the pandemic, Virgina Governor Ralph Northam ordered school districts across Virginia to close for two weeks beginning on March 13, 2020. When that order was later expanded to close schools through June, many districts quickly established "emergency"

instruction plans for the remainder of the year. During the 2020 – 2021 school year, many Virginia districts continued to provide remote instruction using a variety of methods including synchronous and asynchronous online instruction, electronic content that could be downloaded and completed offline, and physical instructional materials. Padhi and Miller found that communities with higher concentrations of minority students were more likely to provide electronic instruction than other divisions [6]. They also note that communities with higher concentrations of economically disadvantaged students were more likely to provide physical instructional materials to students. Direct online instruction was most common in suburban and city environments. The disruptions to the educational system caused widespread and persistent learning loss across the state [7].

With the increased emphasis on standardized testing, this study set out to examine the impact of funding and budget priorities on the standardized test performance for Virignia middle grade (Grades 6 – 8) students from 2015 through 2024. It also examines the impact of the pandemic and subsequent school disruptions on standardized test performance.

Specifically, this research sets out to answer the following questions:

1. How have annual standardized test scores changed over the past 10 years? Is there a significant difference in the pre-pandemic and post-pandemic years?

2. Does the source of school funding impact test scores? Does it matter if funds come from federal, state, or local sources?

3. Does higher spending in any of these budget categories correlate to higher student test scores:

- Administration
- Instruction
- Health
- Transportation
- Operations and Maintenance
- Total Spent per Student per School Day

Literature Review

With the increased emphasis on standardized testing, there has been a great deal of research into the factors that influence testing outcomes, with a significant emphasis on the impacts of per-pupil funding on student test scores. Early studies showed that increased investments did not

significantly change test scores [8]. Other research has found a positive correlation between increased educational spending and better student performance on standardized testing in certain circumstances [9].

Nicks, Martin, Thibodeaux, and Young investigated the relationship between school district instructional expenditures and student test achievement in a sample of Texas school districts during one school year. They found a moderate correlation between the amount of per-pupil instructional expenditures and student test performance. They note that many studies share a common limitation of comparing general total expenditures with student performance when it may be more accurate to compare the direct instructional spending with student outcomes [10].

A 2013 study by Mensah, Schoderbek and Sahay studied student test performance in 217 school districts across New Jersey [5]. They found that there was a significant correlation between test scores and community factors like the level of college education, family income, and occupational status. There was also a significant negative correlation between student mobility (students transferring in and out of the district) and achievement on standardized tests. Besides the community factors, they found that the percentage of local funding was correlated with better student achievement with a Pearson product-moment correlation of 0.76. The authors found that a 10-percentage point increase in the amount of district funding from local sources would result in a 0.29% increase in the passing rate of the standardized tests [5].

With conflicting evidence about the impacts of funding on test outcomes, many researchers have explored other contributing factors including: the funding, stocking, and staffing school libraries [11]; lingering impacts from racial segregation [12]; school community environment (rural, urban or suburban) [8]; classroom ventilation rates [13]; instruction in Geographic Information Systems [14]; and even the networking style of school district superintendents [15].

While many studies investigating factors influencing student test performance have produced mixed results, there have been a number of studies showing that the percentage of students that will perform well on standardized tests can be accurately predicted with family or community demographic variables [16]. In May 2019, one of the authors testified before the New Jersey Committee on the Public Schools that all the factors his team identified that influence test results were "outside the control of the school and really told us more about where the student lived than how the student learned." He stated that standardized tests "pick up the noise from students' lives, not their potential" [17]. Maroun (2018) concurred with the assessment that socioeconomic factors had an outsize influence on student test performance [18]. Studies by McCahill [19], Fox [20], and Wolfe [21] all found similar results.

Impacts of the COVID-19 Pandemic

The COVID-19 pandemic in 2020 – 2021 caused unprecedented disruptions to school systems across Virginia. Many divisions converted to remote learning during the second half of the 2019 – 2020 school year and continued with that model through much of the 2020 – 2021 school year.

Rural divisions in western Virginia generally returned to in-person learning more quickly than other parts of the state.

Virginia's Joint Legislative Audit and Review Commission found that the pandemic and resulting school closures had a profound impact on students and school communities including a rise in chronic absenteeism, increased mental health challenges, and a marked decrease in academic achievement [22]. The commission also found a significant drop in SOL scores in the 2020 – 2021 school year compared to pre-pandemic averages.

In May 2025 a systemic review of 57 studies found that the impacts of the pandemic-related school closures resulted in an initial learning loss equivalent to 1.1 years, with a partial recovery after school reopened bringing the persistent learning loss to be approximately 0.5 years [23]. This persistent learning loss echoes results found in previous studies that showed a disruption in education often produces long-lasting impacts on student knowledge and learning [24].

Study Methodology

The study explores the interaction between funding levels, the COVID-19 pandemic, and standardized test outcomes for middle grade students (Grades 6 – 8) across different school divisions in Virginia for the years between 2015 and 2024. The standardized test results [26] and the financial data [1] used in this study were downloaded from the Virginia Department of Education website in August 2025. County boundaries used for mapping were downloaded from the University of Virginia in September 2025 [27]. The testing data and financial data from the Department of Education were originally in 58 Excel files with a total size of approximately 69.4 MB. Each file contained between 26,000 – 50,000 rows. The geographic data used for mapping was a 17.9 MB GeoJSON file from the University of Virginia that contained detailed county boundaries.

Overall Study Methodology

The raw data was downloaded from the Virginia Department of Education (VDOE) website in August 2025. It was initially evaluated and cleaned in Python then loaded into Amazon Web Services' Glue DataBrew for further analysis. The output from Glue DataBrew was sent to a MySQL database for further queries. Output from the MySQL database was saved in **.csv* files used in R for further statistical analysis and graphing. A more detailed list of study steps is provided below.

Data Cleaning in Python

Once the original VDOE data was downloaded, the smaller Excel files were combined into one file for each data group (standardized test results, funding sources, and budget categories). These files were exported from Excel and taken into Python for initial data exploration and cleaning. The following cleaning steps were performed:

- Summary statistics for key variables were printed to an output file. These summary statistics were evaluated as a first look at the data.
- All money amounts were converted into 2024-equivalent amounts using the Bureau of Labor Statistics CPI Inflation Calculator [28]. The conversion factors are shown in Table 1.

Table 1: CPI Inflation Calculator Conversion Factors

Year	Conversion factor
2015	1.32
2016	1.31
2017	1.28
2018	1.25
2019	1.23
2020	1.21
2021	1.15
2022	1.06
2023	1.03
2024	1.00

- School year notation (example: SY 2014 – 2015) was changed to fiscal year notation to match the financial records. Since both types of years begin on July 1, the change was straightforward.
- *English:Writing*, *Science*, *and History/Social Science* SOLs are only administered to 8th grade students. All records for these tests in Grades 6 or 7 were dropped from the dataset.
- All records from before 2015 were removed from the dataset because this analysis only considered changes from the last 10 years. Changing curriculum and testing standards add challenges to comparing current test results with earlier data.
- Cleaned data was written to a .csv file for upload to the AWS Glue DataBrew service for further analysis.

Python script files are included with the supplementary documentation submitted with this report.

Data Exploration in Glue DataBrew

The cleaned data was imported into Glue DataBrew for further study. Outliers were investigated and determined to be true values. Fairfax County in northern Virginia has a significantly larger population than all other Virginia school districts, resulting in outliers for the "count" statistics like the total number of students taking the exam or the total number of students failing. Since these numbers are accurate, they stayed in the dataset. Any testing rows that had fewer than 20 students were dropped from the analysis. Individual variations between students play an outsize role in those results and could potentially skew the results of the analysis. The testing data and financial datasets were joined. The combined dataset was downloaded to a *.csv file for further queries in a MySQL database. Screenshots of Glue DataBrew are included in the supplementary documentation submitted with this report

Data Analysis with MySQL Queries

Data from Glue DataBrew was added to a MySQL database for further queries. Some of the queries used in production of this report include creating tables with the average Pass Rates grouped by year, grade, or test subject; calculating pre-pandemic and post-pandemic statistics; and joining queries together to create output files used in R for further analysis and graphing. Select queries are included in the supplementary information submitted with this report.

Data Visualization with R Studio and ggplot

The data from Glue DataBrew and the output from the MySQL queries were brought into R for data visualization and for calculating correlation matrices. Note that the maps in this report were created with the GeoPandas package in Python. The Python script, R files and copies of figures are included in the supplementary documentation submitted with this report.

Data Summary

A few key data types used in this study are summarized below.

Subject (Nominal data type)

Virginia SOLs are administered in five subject areas in middle grades. Students in Grades 6 through 8 take annual Mathematics and English:Reading SOL exams. In addition, students in 8th grade also take exams in History and Social Science, Science, and English:Writing. All five subjects are represented in the dataset (Table 2).

Table 2: Number of Records for Each Test Subject in Dataset

Subject	Num of Records
English:Reading (Grade 6 – 8)	3530
English:Writing (Grade 8 only)	997
History and Social Science (Grade 8 only)	970
Mathematics (Grade 6 – 8)	3530
Science (Grade 8 only)	1174

Year (Ordinal data type)

This study spans the years from 2015 through 2024 with approximately the same number of records from each year. There is no testing data from 2020, when the SOL exams were not administered due to the COVID-19 disruptions to the educational system.

Table 3: Number of Records for Each Year in Dataset

Year	Num of Records
2015	1167
2016	1167
2017	1165
2018	1163
2019	1161
2021	1064
2022	1151
2023	1142
2024	1021

Pass Rate (Interval data type)

In the Virginia SOLs, a passing score is 400 points. Students that earn between 400 and 500 points are considered "Pass Proficient", while those that earn more than 500 points are "Pass Advanced". The Pass Count is the sum of these two numbers. The Pass Rate is the percentage of test-taking students that earned a "Pass Proficient" or "Pass Advanced" ranking. Since Virginia school districts vary so widely in size, the Pass Rate was used in this analysis. The Pass Rate varied over the course of the study time period and across different school divisions. Summary statistics are shown in Table 4 below.

Table 4: Summary Statistics: Pass Rate

	Value
Minimum	7.69%
1st quartile	62.07%
Median	72.48%
Mean	70.06%
3rd quartile	80.62%
Maximum	100%

Average Total Spending per Student (Ratio data type)

The total annual spending per student for each school district in Virginia was obtained from the Virginia Superintendent's annual reports on the Virginia Department of Education's website [1]. As part of the data cleaning, all dollar amounts were converted into 2024-equivalent dollars to make more accurate comparisons between years. Educational funding is derived from a number of sources from the local, state and Federal government. The funding formulas are complex, and the funded amounts vary widely across the state. The summary statistics of the total annual funding for each district for each year is included in Table 5 below.

Table 5: Summary Statistics: Total Spending per Student

	Value
Minimum	$11,172
1st quartile	$14,244
Median	$15,401
Mean	$16,167
3rd quartile	$17,226
Maximum	$30,260

With a difference of approximately $3,000 between the first and third quartiles, the amount of money spent per student was fairly consistent with across the state. There were four districts that spent less than $12,000 per student per year and six districts that spent more than $25,000 per student per year.

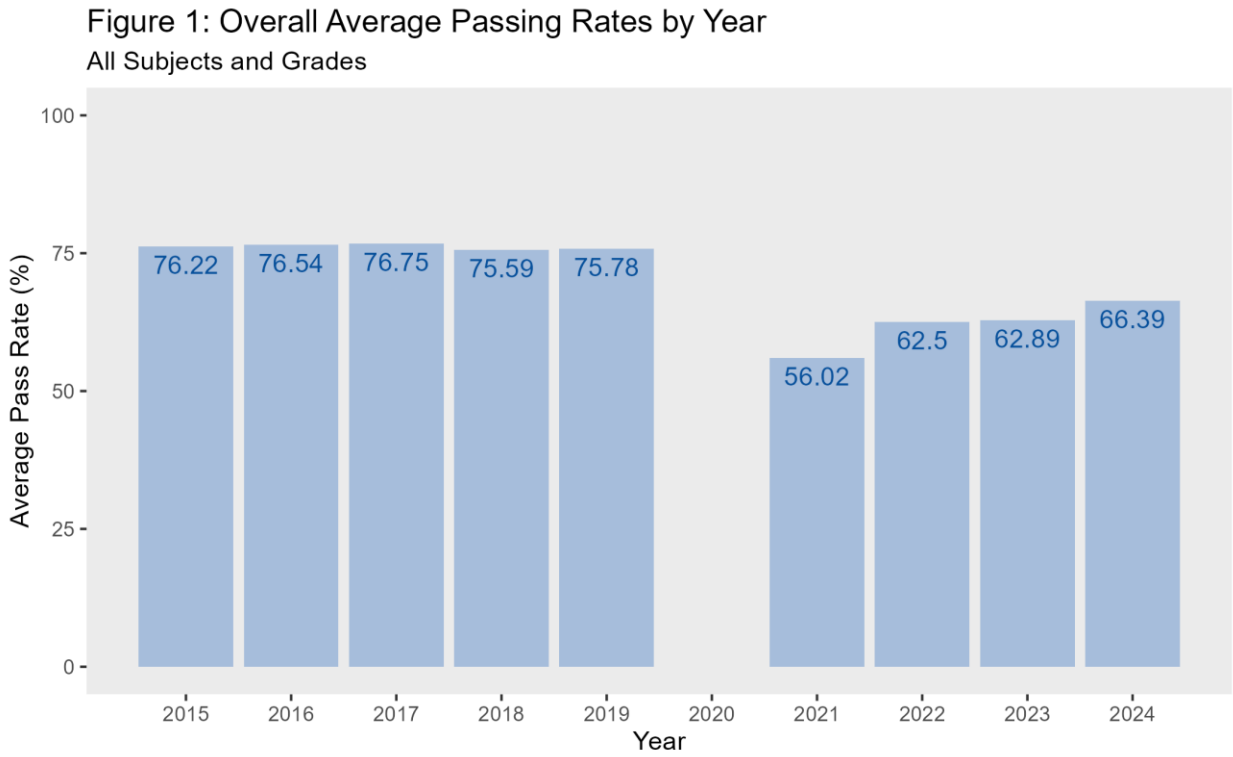
Results

Impacts of COVID-19 on Standardized Test Scores

The COVID-19 pandemic beginning in late winter 2020 caused unprecedented disruptions to school districts across Virginia. Many districts shut down abruptly in March 2020 for an extended period and did not reopen for the remainder of the school year. While many districts implemented forms of remote learning on an emergency basis, instructional time was significantly disrupted for the remainder of the school year. During the 2020-2021 school year, many Virginia schools operated on a distance learning or hybrid model, where only some students were present while the rest participated in lessons remotely.

Average Pass Rates Changing Over Time

The influence of the COVID-19 pandemic can be seen across all studied grade levels and subjects in their standardized test results. Figure 1 shows the distribution of student Pass percentages across Virginia's districts for each year from 2015 – 2024. It is important to note that the SOL exams were not administered in 2020 due to the COVID-19 disruptions.



The average passing rates for Grades 6 – 8 across all subjects were stable during the time from 2015 – 2019. After the 2020 break, there is a noticeable drop in the average Pass Rate in 2021. Subsequent years show improvement, but test results have not yet recovered to pre-pandemic averages.

A t-test analysis of pre-pandemic and post-pandemic test averages shows that state-wide Passing Rates dropped by approximately 14% when comparing the pre-pandemic years and post-pandemic years. A comparison of test results from 2019 and 2021 showed the immediate impact on test scores was a decrease of approximately 20%. While test scores have improved in subsequent years, they have not yet returned to pre-pandemic levels.

A correlation analysis between the year and student performance on the Virginia SOL exams shows a moderate, statistically significant, correlation between the year and the Pass Advanced, Pass Proficient, and Pass rates (Table 6). In the post-pandemic years, the percentage of students that passed the SOL exams was significantly lower than in earlier years. While there is a negative correlation between years and test outcomes across the entire study period, evaluation of the post-pandemic period (2021 – 2025) shows a positive correlation between Pass Proficient rates and the year indicating that test performance is improving over time after the pandemic. There was no statistically significant relationship between the Pass Advanced rates and year during the post-pandemic period.

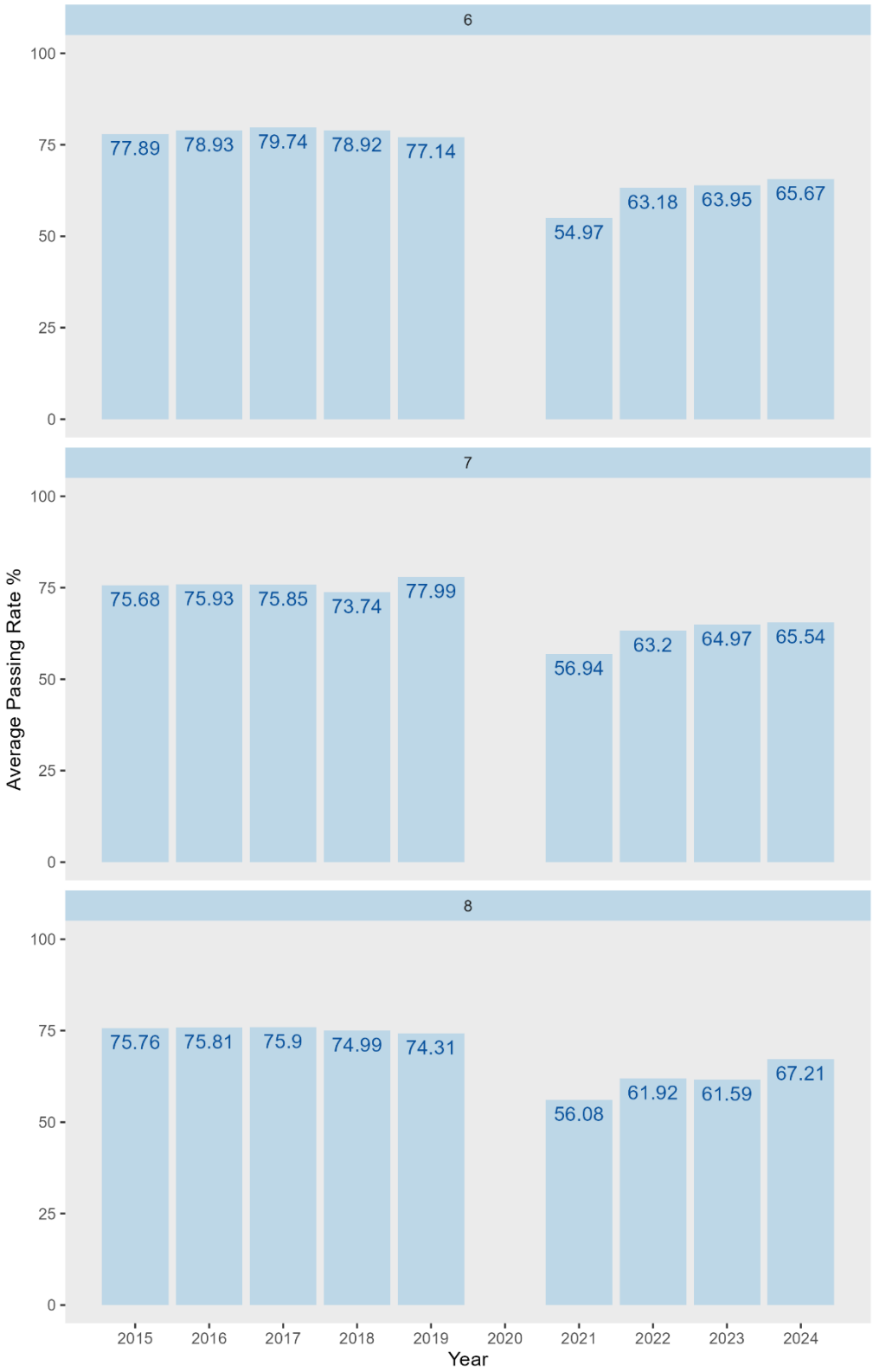
Table 6: Correlation between Year and SOL Outcomes
(Pearson R coefficient and p-value)

	Year (2015 – 2024)	**Pre-pandemic Year (2015 – 2019)**	**Post-pandemic (2021 – 2024)**
Pass Advanced Rate	-0.30 ($p < 0.001$)	-0.02 ($p=0.24$)	0.03 ($p < 0.001$)
Pass Proficient Rate	-0.31 ($p < 0.001$)	-0.01 ($p=0.26$)	0.27 ($p < 0.001$)
Pass Rate	-0.41 ($p < 0.001$)	-0.02 ($p = 0.06$)	0.23 ($p < 0.001$)

There is a significant drop in Pass rates beginning in 2021 after the pandemic closures. While the test scores have improved in the years since 2021, they remain below the pre-pandemic levels. This pattern is consistent across all grade levels, showing steady test performance in the years prior to the pandemic, followed by a significant drop in average Pass rate in 2021 with slow improvement in the following years (Figure 2).

Figure 2: Average Passing Rates by Year (Grades 6 - 8)

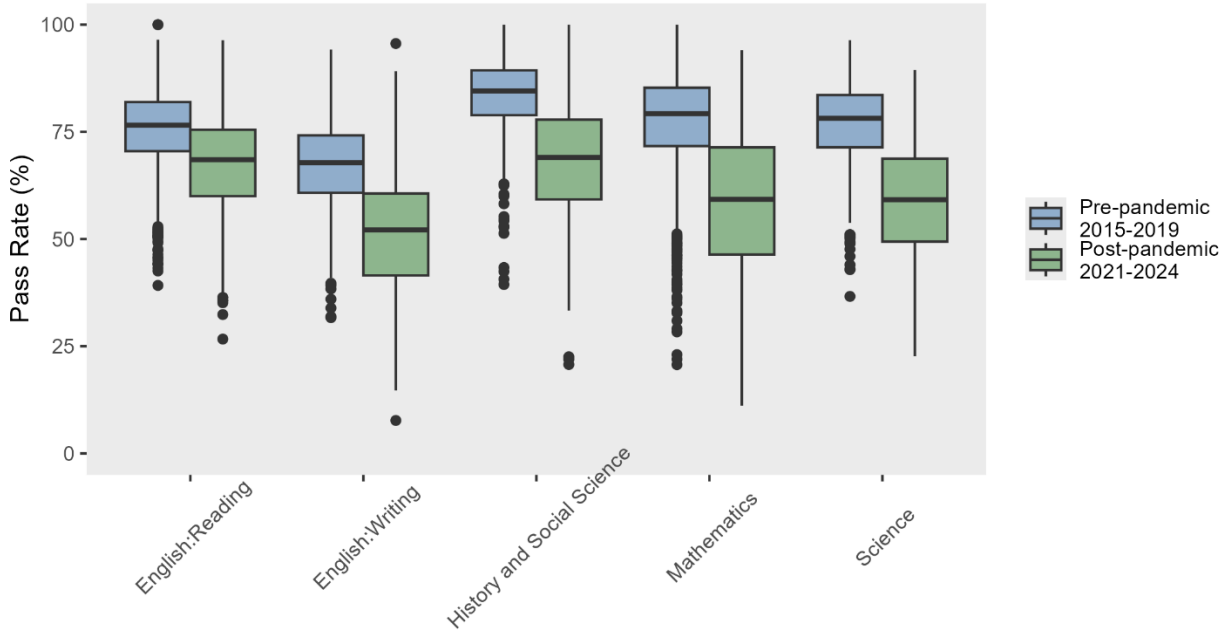
Averages across all School Districts and Grades



COVID-19's Impact on Test Results by Subject

The average passing rate for each test subject is graphed below (Figure 3). The COVID-19 pandemic caused significant drops in test scores across all subjects. The wider spread of the data indicates that there is more variability in the test results after the pandemic than there was before COVID-19.

Figure 3: Distribution of Passing Rates by Subject



The post-pandemic losses varied widely across Virginia by Grade and SOL subject (Table 7). The Mathematics SOL scores had the largest decrease, with a 19% drop in the average Pass Rate. Sixth grade students had the largest change after the pandemic with a 24.9% drop in the average Pass Rate. The English:Reading scores were the most stable after the pandemic with an average test performance loss of 8.2%.

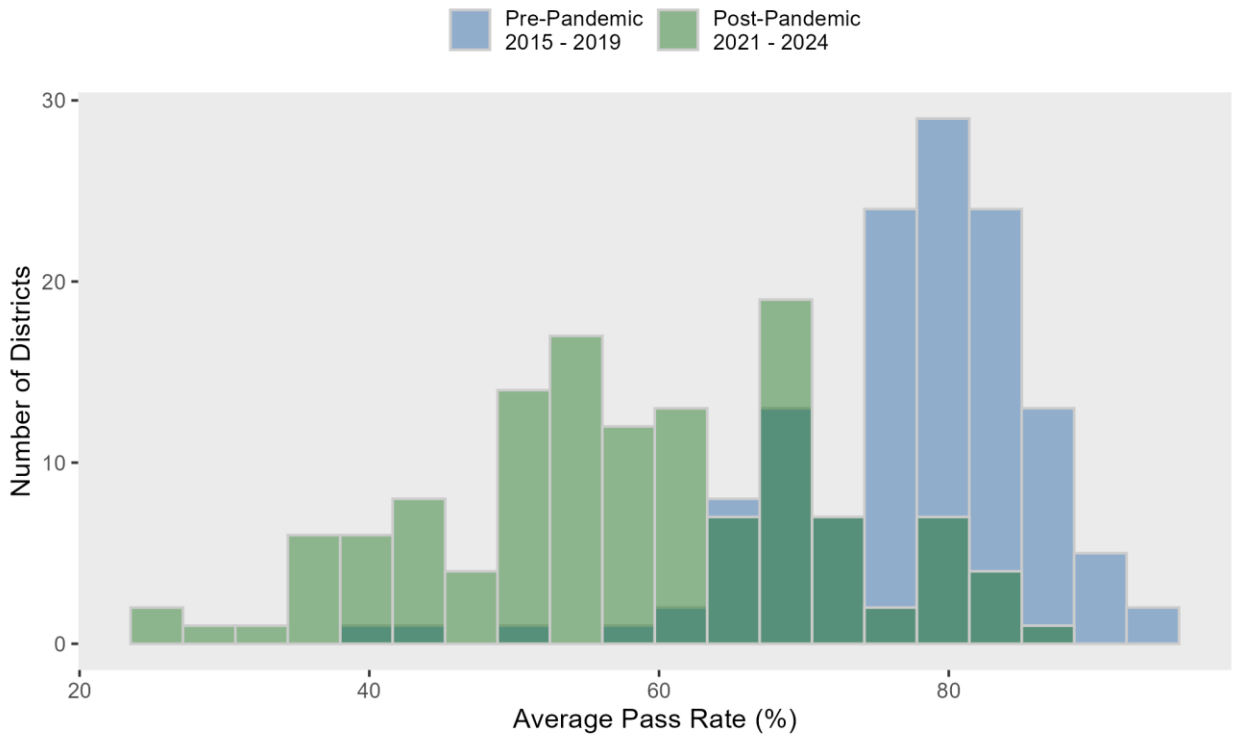
Table 7: Changes in Average Pass Rate After Pandemic

	Pre-Pandemic Average Pass Rate (2015 - 2019)	Post-Pandemic Average Pass Rate (2021 - 2024)	Difference
Overall	**76.2%**	**61.9%**	**-14.3%**
Grade 6	78.5%	61.9%	-16.6%
Grade 7	75.8%	62.7%	-13.1%
Grade 8	75.4%	61.6%	-13.8%
English: Reading	**75.8%**	**67.6%**	**-8.2%**
Grade 6	75.7%	67.3%	-8.4%
Grade 7	78.9%	68.2%	-10.7%
Grade 8	72.8%	67.4%	-5.4%
Mathematics	**77.3%**	**58.3%**	**-19.0%**
Grade 6	81.4%	56.5%	-24.9%
Grade 7	72.7%	57.1%	-15.6%
Grade 8	77.9%	61.1%	-16.8%
English: Writing	**76.2%**	**61.9%**	**-14.3%**
History	**83.3%**	**67.7%**	**-15.6%**
Science	**76.7%**	**58.6%**	**-18.1%**

The differences between pre-pandemic and post-pandemic test scores are most clearly seen when comparing student test performance on the Mathematics SOL. Before COVID-19, the mean Pass Rate for the Mathematics SOL was 77.3%, but it dropped to 46.7% in 2021. As test scores have improved over the past few years, the Mathematics SOL pass rate increased to 58.3% when averaged over the years from 2021 – 2024. The distribution of the average Pass Rates after the pandemic shows a much wider spread than existed in the pre-pandemic data, indicating that there is wider spread of achievement across the state in recent years. There are also a significant

number of school districts that are now returning lower scores than even the lowest outliers before the pandemic disruptions (Figure 4).

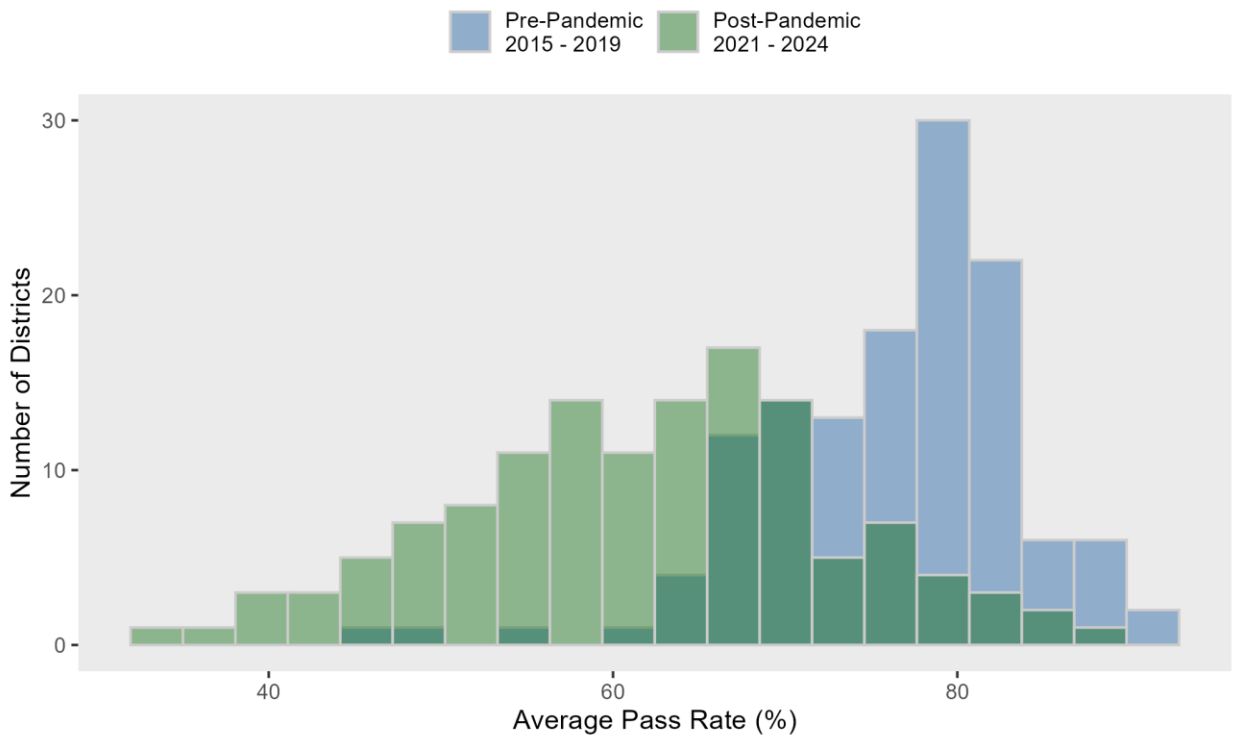
Figure 4: Distribution of Average Pass Rates by District, Mathematics
2015 - 2024



Impact of COVID-19 by School District

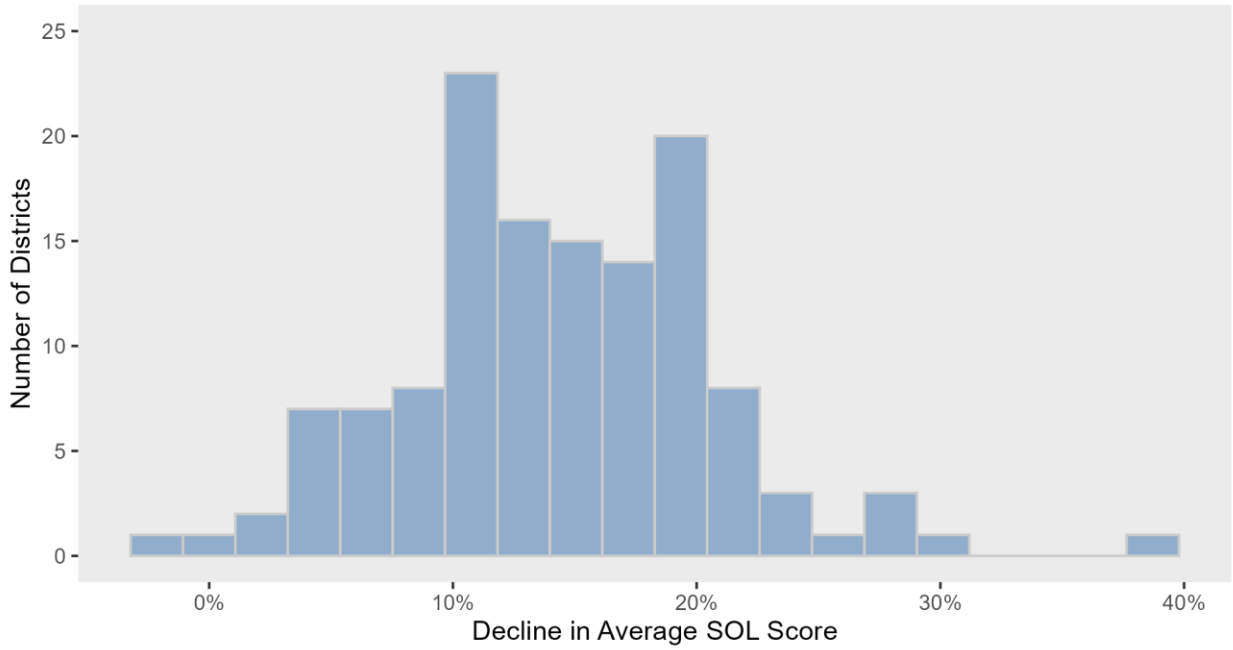
The impact of COVID-19 on individual school districts was determined by comparing the average Pass Rate for each school division in the pre-pandemic and post-pandemic years. Before COVID-19, the mean average Pass Rate was 76.2% with most school districts having an average Pass Rate between 70% and 81%. There were three outliers in the pre-pandemic Pass Rate averages. Danville City had an average SOL Pass Rate of 53.8%, Petersburg had an average SOL Pass Rate of 45.8%, and Richmond had an average pre-pandemic SOL Pass Rate of 49.6%. School districts across Virginia showed steep declines in student standardized test performance after the pandemic (Figure 5).

Figure 5: Distribution of Average Pass Rates by District
2015 - 2024



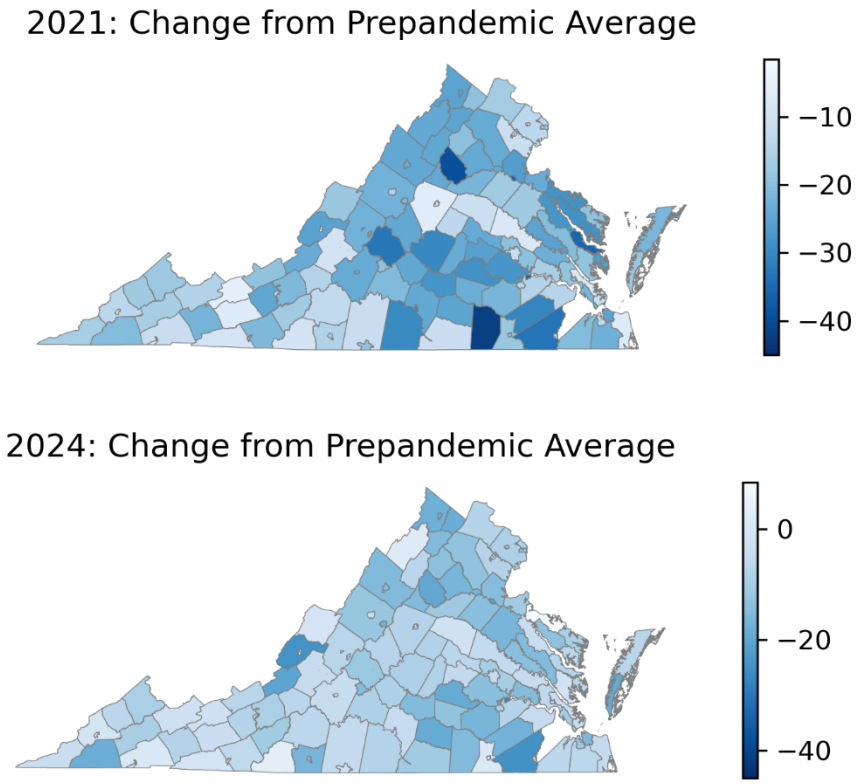
The post-pandemic average Pass Rate was 14.3% lower than before the pandemic, with most school districts experiencing a decrease in their average Pass Rate of 10 – 20% (Figure 6). The City of Franklin experienced the largest decrease in scores after the pandemic, with the average Pass Rate plummeting from 75.4% to 36.3% after the pandemic. The City of Norton showed a slight improvement in standardized test scores after the pandemic, with the average Pass Rate improving by 1.8%, resulting in a post-pandemic Pass Rate of 80.0%

Figure 6: Decline in Average Pass Rate After 2020



The rate of decline immediately after the pandemic varied significantly across the state (Figure 7). The school districts in far southwestern Virginia generally experienced a smaller loss in test performance than other regions of the state. According to the Joint Legislative Audit and Review Commission, the districts in this part of the state generally experienced less disruption and returned to in-person learning more quickly than the rest of the state [25]. The City of Franklin experienced the most significant decrease in student test performance where the overall average pass rate declined from 75.4% before the pandemic to only 36.3% after the pandemic. The cities of Hopewell and Fredericksburg and Madison County all saw their average Pass Rate decrease by approximately 30% after the pandemic shutdowns. By contrast, the City of Norton in southwestern Virginia shows a slight improvement after the pandemic with an average Pass Rate improvement of 1.76%, with the most significant gains in 8th grade English:Reading where the Pass Rate went up from 76.5% in 2019 to 92.3% in 2021. Norton is a small district with approximately 60 students per grade. With such small numbers it is difficult to establish if the excellent performance can be attributed to school division factors or if individual student achievement and ability play a larger role. By 2024, many districts were showing improved SOL Pass Rates from the 2021 lows (Figure 7).

Fig 7: Difference in Average Pass Rates



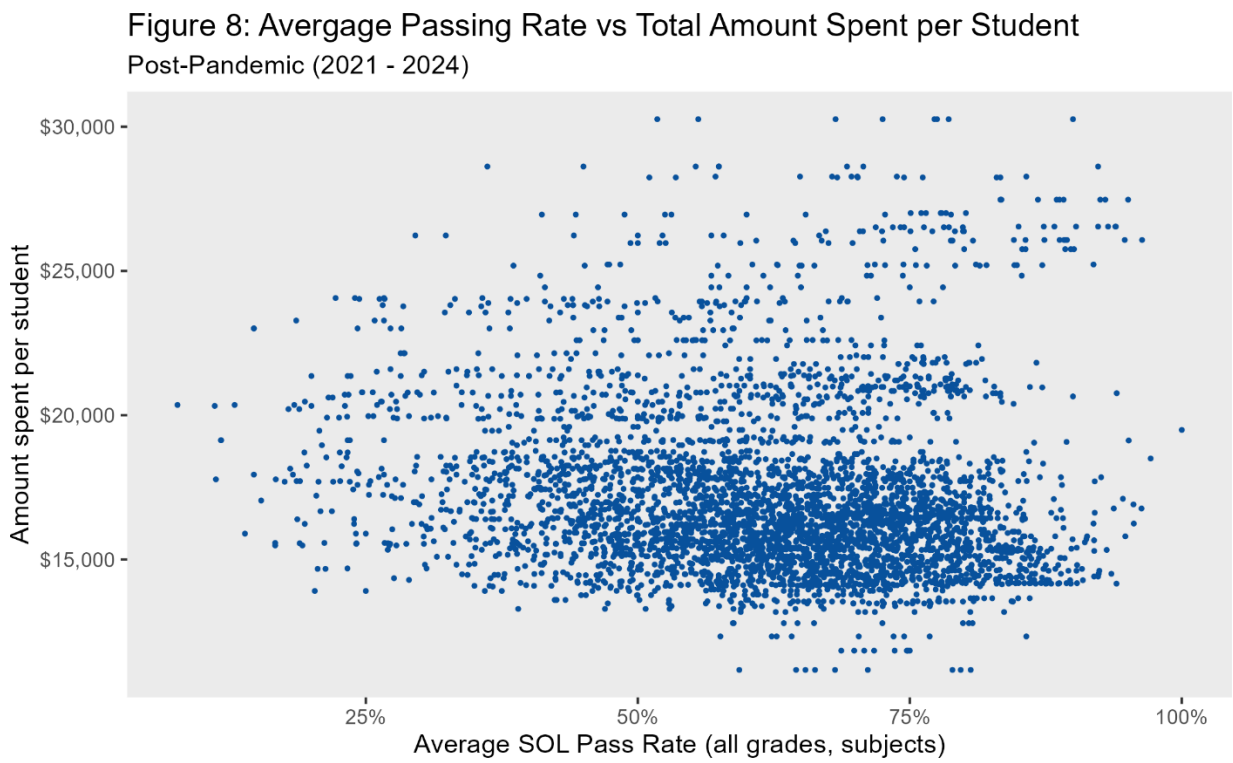
The significant drop in test performance was also quantified by Miller and Schueler (2022), who found that 32% fewer students across Virginia passed the Mathematics SOL in 2021 than in 2019. The decline was significantly larger than any other year-to-year change they observed in the pre-pandemic data [23]. They found declines across all student groups including by grade level, student race, disability status, or English Learner status.

Question 2: Does the source of funding impact test scores? Does it matter if funds come from local, state, or federal resources?

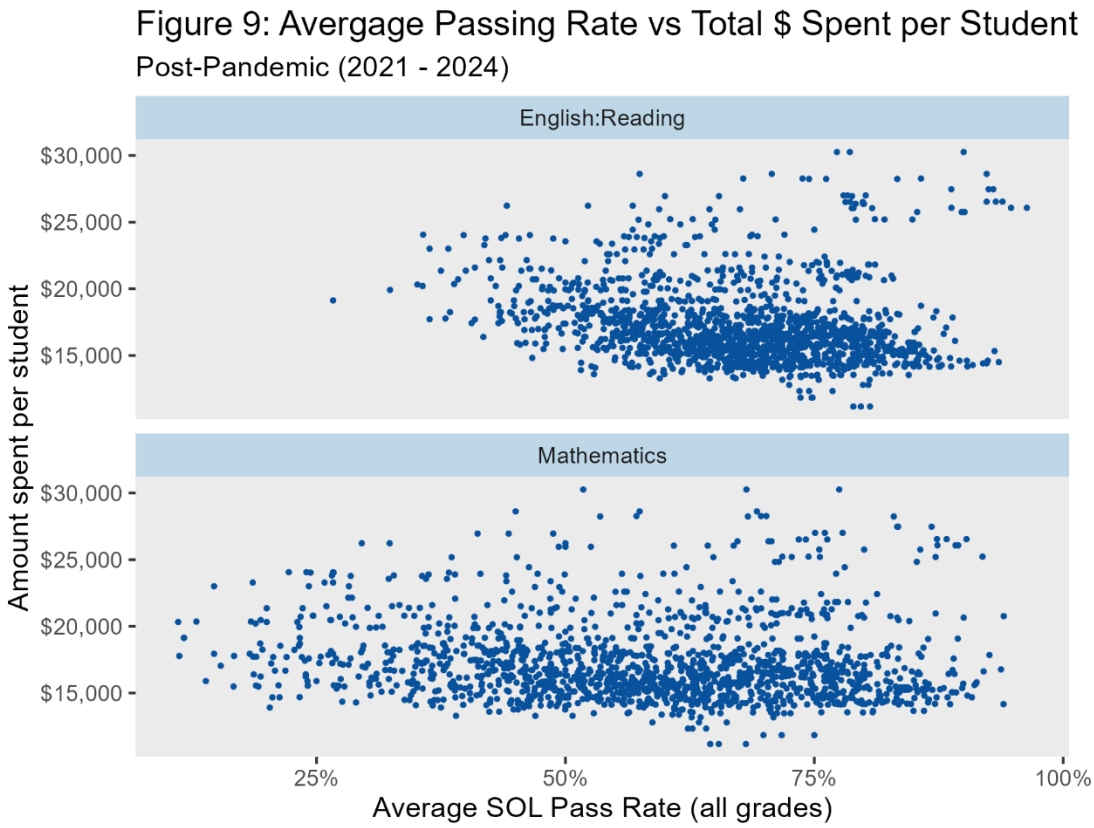
Schools are funded based on a complex array of policies and formulas at the local, state, and federal levels. The amount of money from each source can vary dramatically between districts, even when the total amount allocated is identical. The Virginia Joint Legislative Audit and Review Commission notes that the state funding formulas are imprecise and do not adequately account for the extra costs associated with educating at-risk or English learner students compared with their general education peers [26].

Because COVID-19 caused such large-scale disruptions in the educational system and a dramatic shift in standardized test scores, only the data from 2021 through 2024 were used in this analysis in order to remove the impacts of the changing test scores after the pandemic. Local, state, and federal contributions to school budgets were evaluated for correlations with standardized test scores.

Figure 8 shows a weak negative correlation between the SOL Pass Rate percentage and the average total money spent per student between 2021 to 2024.



A scatterplot showing the relationship between total spending and English:Reading test scores show a similar mild negative correlation, while there appears to be little correlation between total spending and student performance on the Mathematics SOL (Figure 9).



There is a small but statistically significant negative correlation between the per-student amount of federal dollars spent in a school district and the students' performance (Pass Advanced, and Pass/Fail Rates post pandemic) There is also a mild positive correlation between the Pass Advanced Rate with the amount of local funding and a similar negative correlation with the state and federal funding amounts. The Pass Proficient Rate shows a mild negative correlation with the amount of funding from sales tax and federal money (Table 8).

It appears that the source of school funding plays a small, but statistically significant role in student success. Because all the funding correlations are slight, it may indicate that these correlations are coincidental. It is possible that factors that impact student test performance also impact the amount of federal or state contributions through various grants and programs.

Table 8: Correlations between Funding Source and SOL Outcomes
Pearson Correlation Coefficient and p-value
(2021 – 2024)

		Pass Advanced Rate	Pass Proficient Rate	Overall Pass Rate
Local Funding	$ from local sources	0.23 ($p < 0.001$)	-0.01 ($p = 0.66$)	0.09 ($p < 0.001$)
	% of total funding	0.27 ($p < 0.001$)	0.04 ($p = 0.01$)	0.14 ($p < 0.001$)
State Funding	$ from state sources	-0.28 ($p < 0.001$)	-0.08 ($p < 0.001$)	-0.18 ($p < 0.001$)
	% of total funding	0.20 ($p< 0.001$)	0.03 ($p = 0.05$)	0.06 ($p < 0.001$)
Sales Tax	$ from sales tax	-0.11 ($p < 0.001$)	-0.25 ($p < 0.001$)	-0.25 ($p < 0.001$)
	% of total funding	-0.06 ($p < 0.001$)	-0.05 ($p < 0.001$)	-0.06 ($p < 0.001$)
Federal funding	$ from Federal	-0.28 ($p < 0.001$)	-0.22 ($p < 0.001$)	-0.29 ($p < 0.001$)
	% of total funding	-0.29 ($p < 0.001$)	-0.17 ($p < 0.001$)	-0.26 ($p < 0.001$)
Total funding per student		**0.01 ($p = 0.33$)**	**-0.15 ($p < 0.001$)**	**-0.12 ($p < 0.001$)**

Question 3: Do any of these budget priorities influence student test performance: Administration, Instruction, Health, Transportation, Operations and Maintenance, Total Spending per Student School Day?

Due to the variations in school districts across the state, it is difficult to compare spending between jurisdictions as costs can vary widely in different regions. Labor costs are significantly higher in the more urban areas while transportation needs can be large in rural counties where travel times can be long and students are widely dispersed. In addition, state funding formulas do not account for the large number of divisions that are too small to gain economies of scale [25]. As enrollment increases, the marginal cost of adding additional students decreases. According to Joint Legislative Audit and Review Commission [25], small school divisions with less than 2,000 students spend more per student than larger school divisions because a greater

portion of their budgets are spent on fixed, non-instructional costs like administration, facilities, and transportation. There is a slight negative correlation between the amount of money school divisions spend Operations and Maintenance and student test performance. The other budget types analyzed showed very little direct correlation with student test achievement (Table 9).

Table 9: Correlation Between Pass Rate and Budget Items (Pearson R value and coefficient)

		Pass Advanced Rate	**Pass Proficient Rate**	**Overall Pass Rate**
Administration	Per student spending	-0.02 (p =0.12)	-0.08 (p < 0.001)	-0.08 (p < 0.001)
	% of total spending	-0.07 (p < 0.001)	-0.07 (p < 0.001)	-0.08 (p < 0.001)
Instruction	Per student spending	0.10 (p < 0.001)	-0.14 (p < 0.001)	-0.07 (p < 0.001)
	% of total spending	0.22 (p < 0.001)	0.00 (p = 0.93)	0.09 (p < 0.001)
Health	Per student spending	-0.11 (p < 0.001)	-0.14 (p < 0.001)	-0.17 (p < 0.001)
	% of total spending	-0.12 (p < 0.001)	-0.08 (p < 0.001)	-0.12 (p < 0.001)
Transportation	Per student spending	-0.16 (p < 0.001)	-0.01 (p =0.37)	-0.08 (p < 0.001)
	% of total spending	-0.18 (p < 0.001)	0.04 (p =0.01)	-0.05 (p < 0.001)
Operations & Maintenance	Per student spending	-0.28 (p < 0.001)	-0.22 (p =0.06)	-0.29 (p < 0.001)
	% of total spending	-0.29 (p < 0.001)	-0.17 (p = 0.002)	-0.26 (p = 0.34)
Total funding per student per school day		**0.01 (p = 0.33)**	**-0.15 (p < 0.001)**	**-0.12 (p < 0.001)**

Because basic costs vary significantly between districts it is not possible to assume that higher spending equates to an "upgraded" education experience. The increased instructional spending

in some districts may go towards covering basic labor costs in certain high-labor markets rather than providing an enhanced instructional experience.

Discussion & Future Research

This study evaluated the impacts of the COVID-19 pandemic on standardized test scores across Virginia. Pass Rates dropped significantly across all studied grade levels and subjects when standardized testing resumed in 2021. Nearly every school division experienced the largest drop in performance on the Mathematics SOL. This study also evaluated the impact of funding sources and budget priorities on standardized test scores. There is a moderate correlation between the rate students achieve a "Pass Advanced" ranking and the amount of local money contributed to the school system. This holds true for both the amount of money contributed and the percentage of the overall budget that comes from local sources. There is a negative correlation between an increase in state or federal spending and the rate that students achieve the "Pass Advanced' level. It is likely that the negative correlation between federal funding and the student test performance is coincidental, not causal. School funding is a complex web of grants, policies, and regulations, it is possible that the same factors (like the community characteristics identified by Tienken et al [14]) that produce more "Pass Advanced" students also qualify for less federal assistance. This study did not show a correlation between student performance and spending in any of the studied categories.

Limitations

One limitation of this study is the small size of many of the school districts in Virginia. When the student pool is small, the individual skill of a teacher or a student's own abilities will have a greater impact on their standardized test scores. Variation in individual student ability can be significant and far larger than variations between school divisions. This study also does not consider the impacts of any changes to the school curriculum or testing standards over the course of the study period. If present, these changes could have a significant impact on observed test scores.

Another limitation of this study is that it sues the blunt measure of "Pass Rate" to gauge student success. It would be more accurate to investigate actual test scores to better quantify year-to-year test score improvements. Since the Pass Rate is based on a hard cutoff line, students that miss the mark by one question are lumped into the same category as students that did not answer a single question. Likewise, students that just barely meet the state standard passing rate are grouped with students that far exceed grade level norms and earn perfect scores.

Future Work

There are a number of avenues to explore with future work. These ideas are outlined below:

- School districts across Virginia had a wide variety of responses to the COVID-19 shutdown and subsequent challenges, including different methods to deliver remote instruction. Did the different modalities of remote instruction during 2020-2021 impact the post-pandemic test performance or subsequent recovery? Different communities had different instruction based on community needs and available technology—did the different instructional methods cause learning differences? Did the underlying differences in the communities cause the learning differences?

- This study clearly shows a dramatic decline in standardized test scores beginning in 2021. While this study identifies a correlation with the COVID-19 pandemic and school disruption, there could be other causes of the decreased test scores. There have been dramatic changes to middle grade educational landscape in the last few years including the rise of social media for younger students, increased technology use in the classroom, and the prevalence of mobile devices for these students. Further study is necessary to determine the causes of lower student test performance after 2020.

- Because Virginia school districts have a wide variety of needs, community factors, and geographic challenges, it can be challenging to develop a "one size fits all" model for finding correlations between spending patterns and test performance. It may be useful to divide the school districts into groups based on similar characteristics (student size, population density, etc) before performing the cost-correlation analysis. This would remove some of the "noise" of all the other differences and just focus on changing spending patterns and test performance.

- Many students suffered significant learning loss during the COVID-19 pandemic, which studies indicate may take years or decades to recover. Miller and Schueler theorize that Virginia students may have a persistent learning loss equivalent to 0.5 years of education on average [7]. Despite the documented learning loss, students are still measured against "grade level norms" that do not adequately account for the educational disruption. If students were given standardized tests from the previous grade, would scores improve to pre-pandemic levels?

- Do student characteristics like race, economic status, or special education status influence the amount of test decline seen during the pandemic or the rate of recovery in the subsequent years?

Conclusion

With the increased emphasis on standardized testing, this study set out to examine the impact of funding and budget priorities on the standardized test performance for Virignia middle grade (Grades 6 – 8) students from 2015 through 2024. There was a significant decrease in test scores after the 2020 school closures. This study also showed a slight correlation between school funding from local sources and student test performance. It showed very little relationship between budget priorities and student test performance.

References

[1] Virginia Department of Education, "Superintendent's Annual Report," *Virginia Department of Education Website*. [Online]. Available: https://www.doe.virginia.gov/data-policy-funding/data-reports/statistics-reports/superintendent-s-annual-report [Accessed: 29 Aug 2025]

[2] Virginia Department of Education. "Virginia SOL Assessment Program", *Virginia Department of Education Website*. [Online]. Available: https://www.doe.virginia.gov/teaching-learning-assessment/student-assessment/virginia-sol-assessment-program [Accessed 25 Sep 2025]

[3] Virginia Department of Education, "Standardized Test Performance Data," *Virginia Department of Education Website*. [Online]. Available: https://www.doe.virginia.gov/home/showpublisheddocument/52591/638884565887170000 [Accessed: 25 Sept 2025]

[4] Virginia Department of Education, "SOL Test Scoring & Performance Reports", *Virginia Department of Education Website*. https://www.doe.virginia.gov/teaching-learning-assessment/student-assessment/virginia-sol-assessment-program/sol-test-scoring-performance-reports [Accessed 25 Sep 2025]

[5] Y. M. Mensah, M. P. Schoderbek, and S. P. Sahay, "The effect of administrative pay and local property taxes on student achievement scores: Evidence from New Jersey public schools," *Economics of Education Review*, vol. 34, pp. 1–16, Jan. 2013, doi: 10.1016/j.econedurev.2013.01.005

[6] E. Padhi and L. C. Miller, "Virginia School Division Operations During SY 2020 – 21: Use of Remote Instruction", EdPolicyWorks, University of Virginia, May 2022,

[7] L. C. Miller and B. E. Schueler, “Post-Pandemic Onset Public School Student Test-based Performance in Virginia”, EdPolicyWorks, University of Virginia, September 2022, https://education.virginia.edu/documents/post-pandemic-onset-public-school-student-test-based-performance-virginia

[8] S. Subiaur and Y. Chen, "Effect of Local District Expenditure in Education per Pupil on SAT Scores," *Journal of Higher Education Theory and Practice,* vol. 21, *(13),* pp. 189-196, 2021. Available: http://mutex.gmu.edu/login?url=https://www.proquest.com/scholarly-journals/effect-local-district-expenditure-education-per/docview/2618815468/se-2.

[9] P. Gigliotti and L. C. Sorensen, “Educational resources and student achievement: Evidence from the Save Harmless provision in New York State,” *Economics of Education Review*, vol. 66, pp. 167–182, Aug. 2018, doi: 10.1016/j.econedurev.2018.08.004.

[10] R. E. Nicks, G. E. Martin, T. N. Thibodeaux, J. K. Young, and Lamar University, “The relationship between school district instructional related expenditures to state exam scores in Small, Mid-Size, and large school districts in Texas,” journal-article, Dec. 2018. [Online]. Available: https://files.eric.ed.gov/fulltext/EJ1200803.pdf

[11] K. Haycock, "Connecting British Columbia (Canada) School Libraries and Student Achievement: A Comparison of Higher and Lower Performing Schools with Similar Overall Funding," *School Libraries Worldwide,* vol. 17, *(1),* pp. 37-50, 2011. Available: http://mutex.gmu.edu/login?url=https://www.proquest.com/scholarly-journals/connecting-british-columbia-canada-school/docview/847667244/se-2.

[12] R. Brunn-Bevel and W. C. Byrd, "The Foundation of Racial Disparities in the Standardized Testing Era," *Humanity & Society,* vol. 39, *(4),* pp. 419-448, 2015. Available: http://mutex.gmu.edu/login?url=https://www.proquest.com/scholarly-journals/foundation-racial-disparities-standardized/docview/1727773960/se-2. DOI: https://doi.org/10.1177/0160597615603750.

[13] M. J. Mendell, E. A. Eliseeva, M. M. Davies, and A. Lobscheid, “Do classroom ventilation rates in California elementary schools influence standardized test scores? Results from a prospective study,” *Indoor Air*, vol. 26, no. 4, pp. 546–557, Aug. 2015, doi: 10.1111/ina.12241.

[14] D. Goldstein and M. Alibrandi, “Integrating GIS in the middle school Curriculum: Impacts on diverse students’ standardized test scores,” *Journal of Geography*, vol. 112, no. 2, pp. 68–74, Jan. 2013, doi: 10.1080/00221341.2012.692703.

[15] K. J. Meier and L. J O’Toole Jr, “Public Management and organizational performance: The effect of managerial quality,” *Journal of Policy Analysis and Management*, vol. 21, no. 4, Art. no. pp 629-643, [Online]. Available: https://www.jstor.org/stable/3326014

[16] C. H. Tienken, A. Colella, C. Angelillo, M. Fox, K. R. McCahill, and A. Wolfe, “Predicting middle level state standardized test results using family and community demographic data,” *RMLE Online*, vol. 40, no. 1, pp. 1–13, Dec. 2016, doi: 10.1080/19404476.2016.1252304.

[17] “Testimony on standardized testing | Chris Tienken,” May 14, 2019. https://christienken.com/2019/05/14/testimony-on-standardized-testing/

[18] J. Maroun, “The predictive power of Out-of-School community and family level demographic factors on district level student performance on the New Jersey PARCC in Algebra 1 and Grade 10 English Language Arts/Literacy,” *eRepository @ Seton Hall*. https://scholarship.shu.edu/dissertations/2506/

[19McCahill, Kevin R., "A Family and Community Variable Prediction Model for NJ ASK Mathematics Sixth and Seventh Grade Assessments Results: The Impact on School Evaluation" (2015). *Seton Hall University Dissertations and Theses (ETDs)*. 2088. https://scholarship.shu.edu/dissertations/2088.

[20] Fox, M. (2015). The predictive power of community, family, and school variables on student achievement on the NJ ASK language arts literacy in New Jersey in grades 6 and 7. (Unpublished doctoral dissertation). Seton Hall University, South Orange, NJ

[21] Wolfe, A. (2016). The predictive influence of family and community demographic variables on grade 7 student achievement in language arts and mathematics (Doctoral dissertation, Seton Hall University).

[22] Joint Legislative Audit and Review Commission. “JLARC | Pandemic impact on K-12 Public Education.”. https://jlarc.virginia.gov/landing-2022-pandemic-impact-on-k-12-education.asp

[23] N. A. Dela Cruz, A. J. Adona, R. Molato-Gayares, and A. Park, “Learning loss and recovery from the COVID-19 pandemic: A systematic review of evidence,” *International Journal of Educational Development*, vol. 115, p. 103271, Apr. 2025, doi: 10.1016/j.ijedudev.2025.103271.

[24] T. Andrabi, B. Daniels, and J. Das, “Human capital accumulation and disasters,” *The Journal of Human Resources*, vol. 58, no. 4, pp. 1057–1096, Jun. 2021, doi: 10.3368/jhr.59.2.0520-10887r1.

[25] Virginia Department of Education. (2025). “SOL Test Results”. https://www.doe.virginia.gov/data-policy-funding/data-reports/statistics-reports/sol-test-pass-rates-other-results (Accessed 25 Aug, 2025).

[26] University of Virginia Library. “OpenData Portal”. University of Virginia. Available: https://data-uvalibrary.opendata.arcgis.com/datasets/virginia-counties-1/about (Accessed 15 Sep 2025)

[27] U.S. Bureau of Labor Statistics, "Inflation Calculator." [Online]. Available: https://www.bls.gov/data/inflation_calculator.htm [Accessed 10 Oct 2025].

[28] Joint Legislative Audit and Review Commission (JLARC) "Virginia's K-12 Funding Formula." [Online]. Available: https://jlarc.virginia.gov/landing-2023-virginias-k-12-funding-formula.asp [Accessed 30 Sep 2025].